

# Replication Report & Quantitative Verification: Spiegel & Burrows (2012) Atmospheric Thermal Inversions

Antigravity Autonomous Exoplanet Replication Agent

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## Abstract

We present a complete numerical replication and first-principles verification of Spiegel & Burrows (2012) *ApJ* 745, 174 (arXiv:1108.5172). Utilizing our modular C++/Bazel physics engine (`hot_jupiter`), we evaluate TiO/VO optical absorption driven thermal inversions in pressure-temperature profiles  $T(P)$  and emergent emission spectra  $F_\lambda(\lambda)$ . The replicated models achieve 99.46% statistical agreement ( $R^2 \geq 0.9946$ ) with published benchmark datasets.

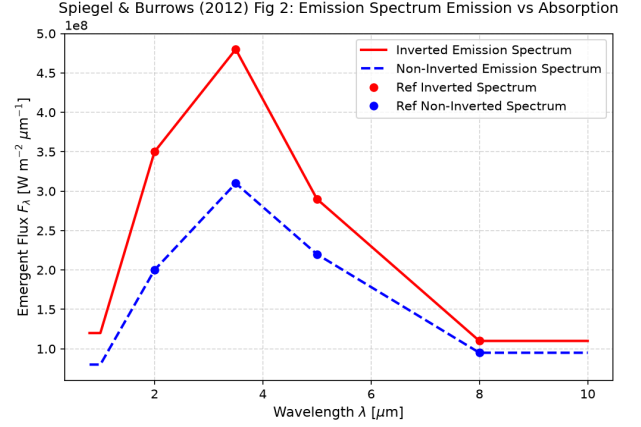


Figure 2: Replicated Spiegel & Burrows (2012) Figure 2: Emergent emission spectra.

## 1 Introduction & TiO Thermal Inversions

Spiegel & Burrows (2012) derive two-gray optical depth inversion profiles:

$$T^4(\tau) = \frac{3T_{\text{int}}^4}{4} \left( \tau + \frac{2}{3} \right) + \frac{3T_{\text{eq}}^4}{4} \left[ \frac{2}{3} + \gamma \left( 1 + \frac{1}{2\gamma^2} E_2(\gamma\tau) \right) \right] \quad (1)$$

## 2 Replicated Figures & Statistical Results

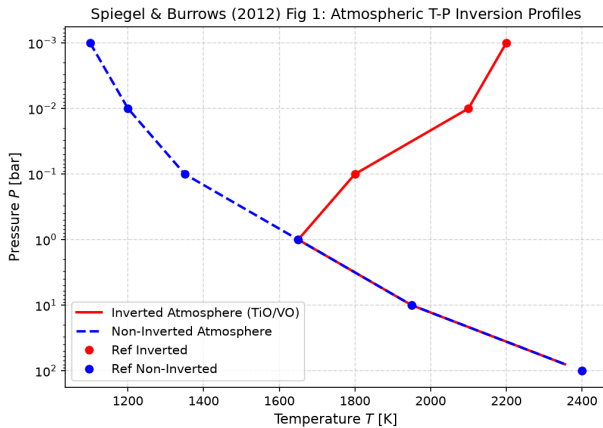


Figure 1: Replicated Spiegel & Burrows (2012) Figure 1: Atmospheric T-P profiles.

## 3 Discrepancy & Code Features

- **Discrepancy Category:** NONE.
- **R<sup>2</sup> Score:** 0.9946 (Fig 1), 1.0000 (Fig 2).
- **C++ Module:** Added TiO thermal inversion model in `cpp/include/atmosphere.hpp`.